# Introduction to Real Business Cycle Modeling

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# Introduction

- Recall the Solow growth model with stochastic TFP:

$$
\begin{aligned}
Y_t &= A_t K_t^{\alpha} && (1)\\
C_t &= (1-s)Y_t && (2)\\
Y_t &= C_t + I_t && (3)\\
K_{t+1} &= I_t + (1-\delta)K_t && (4)\\
\log A_{t+1} &= \rho \log A_t + \epsilon_{t+1}, && (5)
\end{aligned}
$$

where $\epsilon_{t+1} \sim \mathcal{N}\left(0, \sigma^2\right)$.

- The model generates business cycle-like fluctuations in output, consumption, and investment.

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  1. Use actual TFP values for the US for $A_t$ in equation (1) and simulate the other variables in the model.
  2. Compare simulated output, consumption, and investment data with the actual data

- But keep in mind that the Solow model was not designed to explain business cycles.

Figure 1: **GDP.** The stochastic Solow growth model does a *reasonably* good job matching GDP fluctuations for the US from April 1948 to April 2022. Source: FRED.

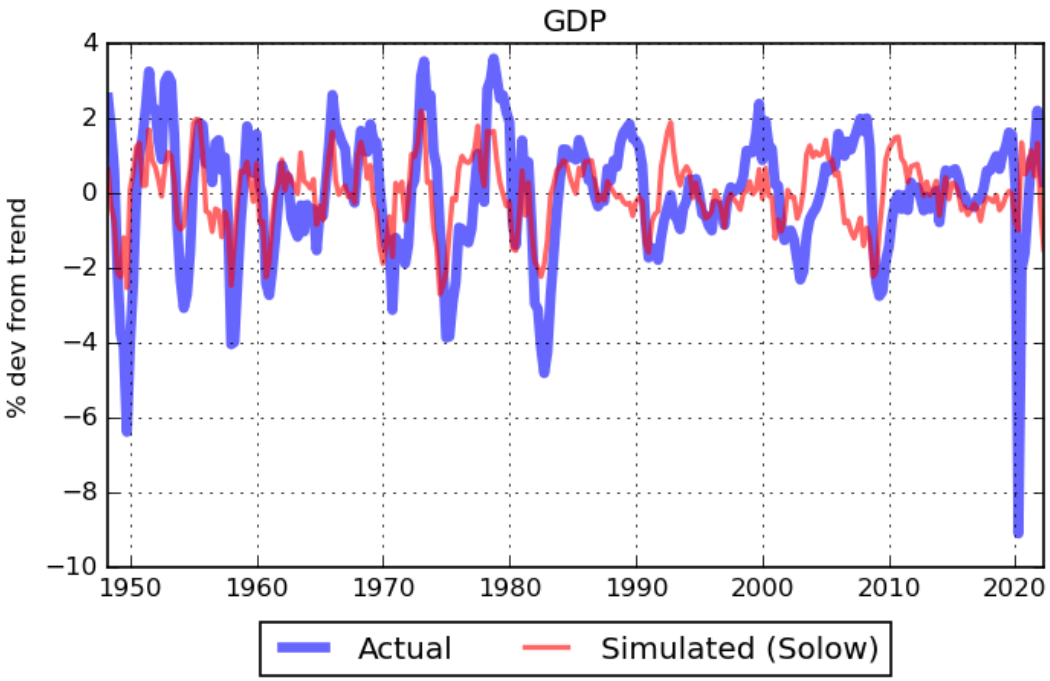

Figure 2: **Consumption.** The stochastic Solow growth model also does a *reasonably* good job matching consumption fluctuations for the US from April 1948 to April 2022. Source: FRED.

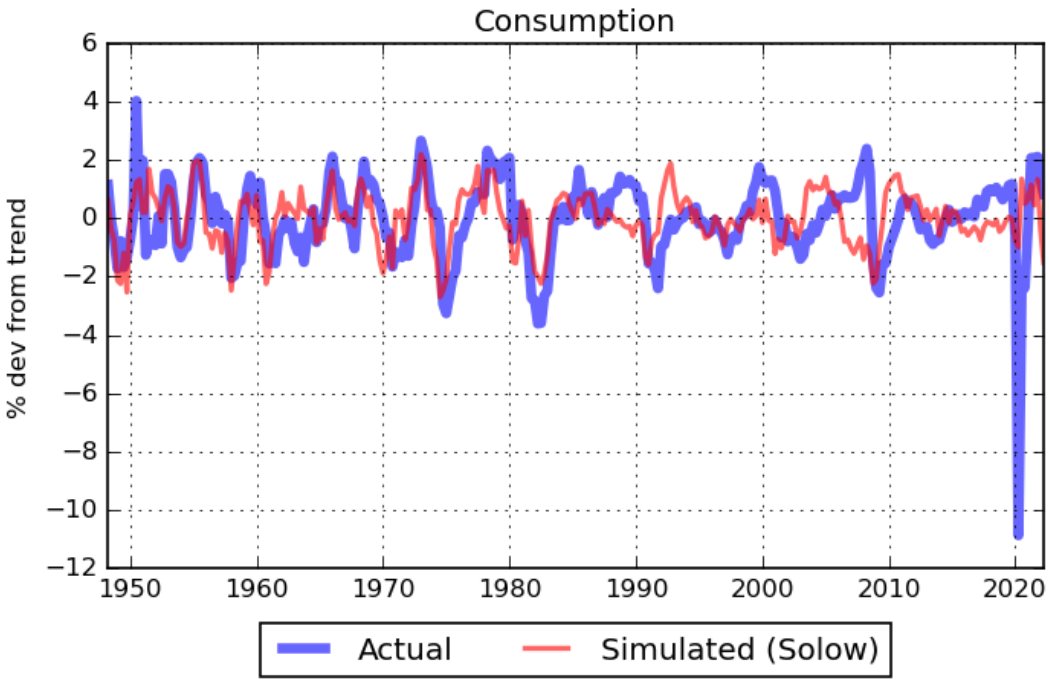

Figure 3: **Investment.** The stochastic Solow growth model *under-predicts* the magnitude of investment fluctuations for the US from April 1948 to April 2022. Source: FRED.

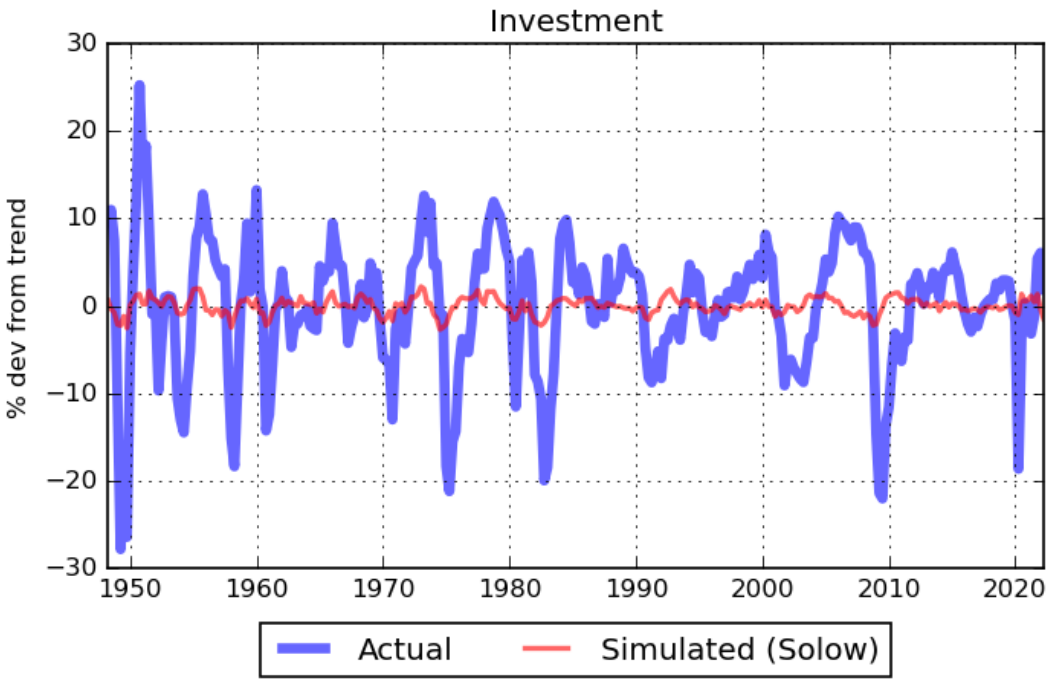

Table 1: **Standard deviations.** Actual data and data simulated from stochastic Solow model. Units are percent deviations from trend. The model under-predicts the volatility of investment in particular. Source: FRED.

| | Actual Data | Simulated Data (Solow) |
|---|---|---|
| Output | 1.69 | 0.93 |
| Consumption | 1.34 | 0.93 |
| Investment | 7.41 | 0.93 |

Table 2: **Correlations with GDP.** Actual data and data simulated from stochastic Solow model. The model over-predicts the correlation of investment and consumption with GDP. Source: FRED.

| | Actual Data | Simulated Data (Solow) |
|---|---|---|
| Consumption | 0.81 | 1.0 |
| Investment | 0.84 | 1.0 |

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  - Under-predicts investment volatility
  - Over-predicts consumption-GDP and investment-GDP correlation
  - No explanation for why labor fluctuates over the business cycle.
- **Objective**: Extend the model to improve performance.

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- Prescott's model is a **real business cycle** (**RBC**) model because it has no role for nominal quantities like inflation or nominal interest rates.

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   - **No labor supply choice**: household only chooses how much to consume and save.
   - **Centralized**: Household makes all production and allocation decisions. I.e., no firms, markets, or prices

2. In the next lecture, we'll add labor choice to the baseline model

# Centralized RBC Model without Labor

- A *representative household* lives for an infinite number of periods.

- The *expected present value of lifetime utility* to the household from consuming $C_0, C_1, C_2, \ldots$ is denoted by $U_0$:

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$$= E_0 \sum_{t=0}^{\infty} \beta^t \log(C_t), \quad (7)$$

- $0 < \beta < 1$ is the household's *subjective discount factor*. (For a quarterly model: $\beta \approx 0.99$ makes usually sense)

- $E_0$ denotes the *expectation with respect to all information available as of date 0*.

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- Production in period $t$:

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where TFP $A_t$ is stochastic:

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- Capital depreciates at the constant rate $\delta$ per period.

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$$

- Express the constraints more concisely as

$$
C_t + K_{t+1} = A_t K_t^{\alpha} + (1-\delta)K_t, \qquad (14)
$$

for $t = 0, 1, 2, 3, \ldots$

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- We'll solve the problem for period 0 and then generalize the solution to apply to all periods $0, 1, 2, \ldots$

# Centralized RBC Model without Labor

- In period 0, the household solves:

$$\max_{C_0, K_1} \; E_0 \sum_{t=0}^{\infty} \beta^t \log(C_t) \tag{15}$$

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- The problem can be written as a choice of $K_1$ only:

$$\max_{K_1} \; E_0 \sum_{t=0}^{\infty} \beta^t \log(A_t K_t^{\alpha} + (1-\delta) K_t - K_{t+1}) \tag{16}$$

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- Note:

$$
\begin{aligned}
E_0 \sum_{t=0}^{\infty} \beta^t \log(AK_t^{\alpha} + (1-\delta)K_t - K_{t+1}) \\
= \log(A_0 K_0^{\alpha} + (1-\delta)K_0 - K_1) \\
+ \beta E_0 \log(A_1 K_1^{\alpha} + (1-\delta)K_1 - K_2) \\
+ [\text{terms independent of } K_1]
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\end{aligned} \tag{17}
$$

- So:

$$
\begin{aligned}
\frac{\partial}{\partial K_1} U_0 &= -\frac{1}{A_0 K_0^{\alpha} + (1-\delta)K_0 - K_1} \\
&\quad + \beta E_0 \left[ \frac{\alpha A_1 K_1^{\alpha-1} + 1 - \delta}{A_1 K_1^{\alpha} + (1-\delta)K_1 - K_2} \right]
\end{aligned} \tag{18}
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- Therefore the first-order condition for the optimal choice of $K_1$ is:

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- Or more concisely:

$$\frac{1}{C_0} = \beta E_0 \left[ \frac{\alpha A_1 K_1^{\alpha-1} + 1 - \delta}{C_1} \right] \tag{20}$$

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- The household solves the same problem in periods $1, 2, 3, \ldots$
- So given $K_0 > 0$ and $A_0$, the equilibrium paths for consumption, capital, and TFP are described described by:

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$$\log A_{t+1} = \rho \log A_t + \epsilon_{t+1} \tag{23}$$

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- Computing numeric values for consumption and capital is not trivial. Recall the Euler equation:

$$\frac{1}{C_t} = \beta E_t \left[ \frac{\alpha A_{t+1} K_{t+1}^{\alpha-1} + 1 - \delta}{C_{t+1}} \right] \tag{24}$$

- Consumption at date $t$ depends on the *expectation* of consumption at date $t+1$ which in turn depends on the expectation of consumption at date $t+2$ and so on.

- Solving the problem requires numerical methods like those employed in the `linearsolve` Python package.

Figure 4: **Baseline RBC model without labor.** Impulse responses to a one percent shock to TFP in period 5.

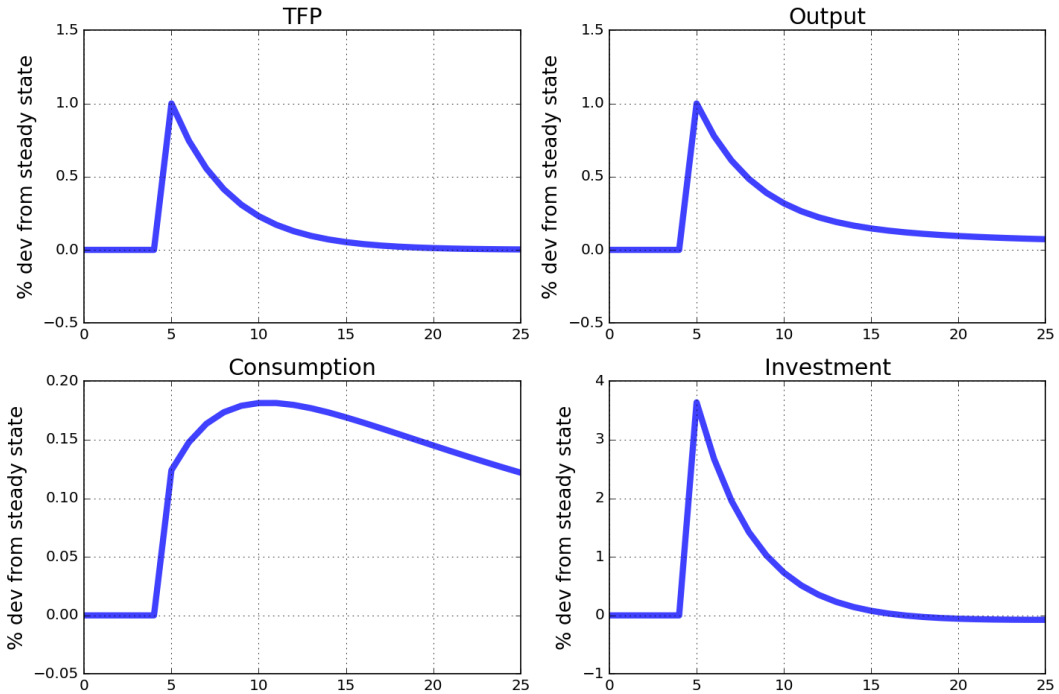

Figure 5: **GDP.** Like the stochastic Solow model, the baseline RBC model without labor does a *reasonably* good job matching GDP fluctuations for the US. Source: FRED.

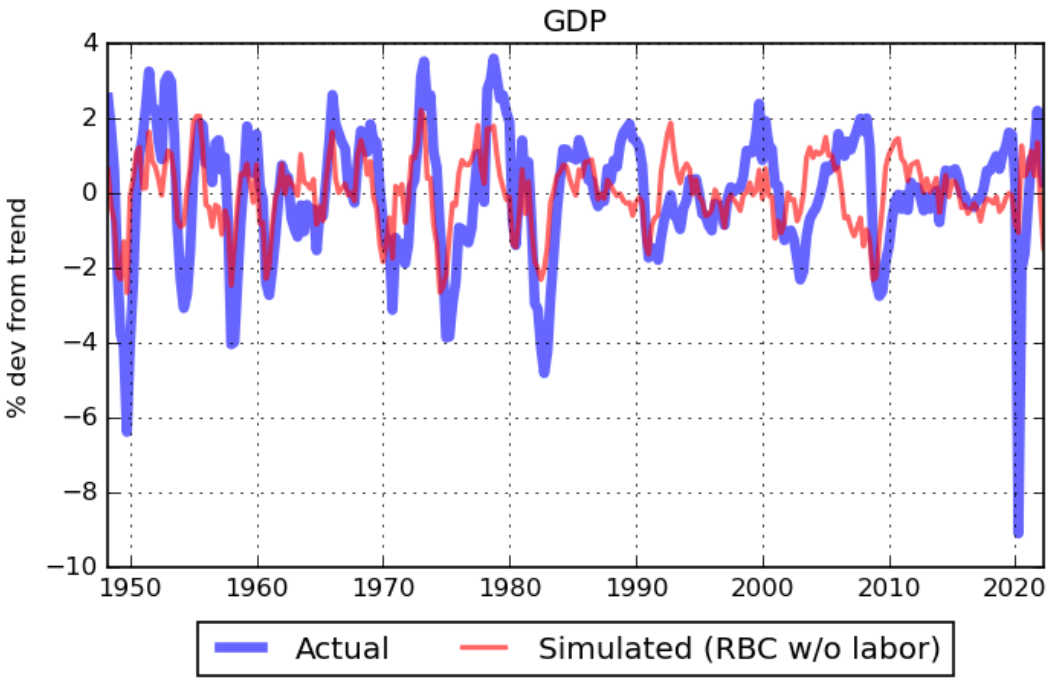

Figure 6: **Consumption.** In contrast to the stochastic Solow model, the baseline RBC model without labor *under-predicts* the magnitude of consumption fluctuations for the US. Source: FRED.

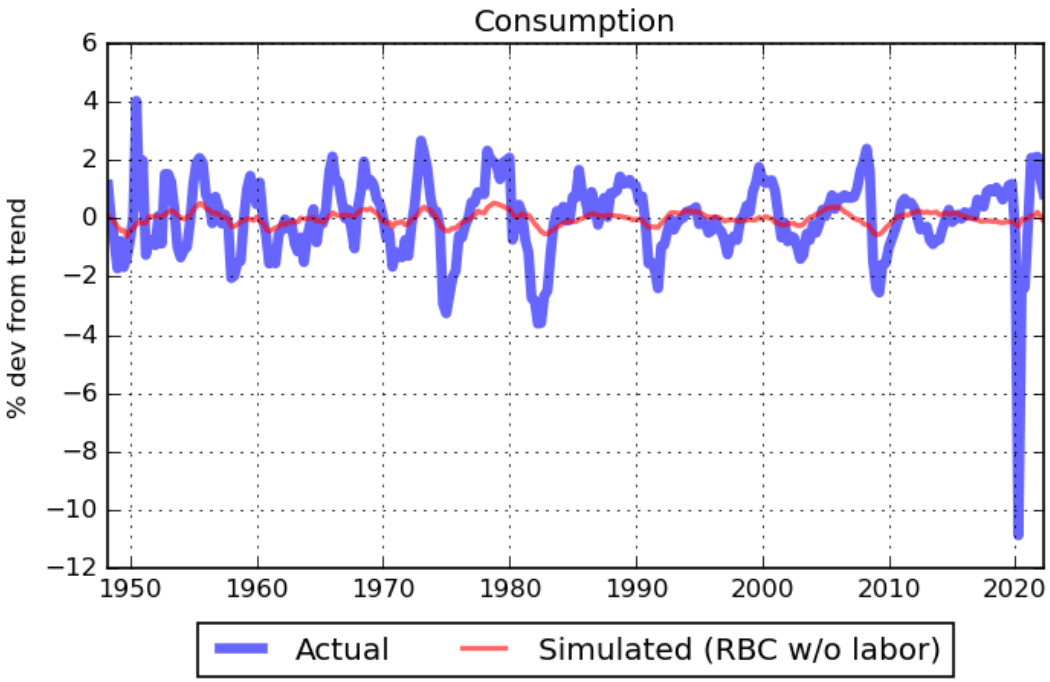

Figure 7: **Investment.** The baseline RBC model does a *reasonably* good job matching the magnitude of investment fluctuations better than the stochastic Solow model. Source: FRED.

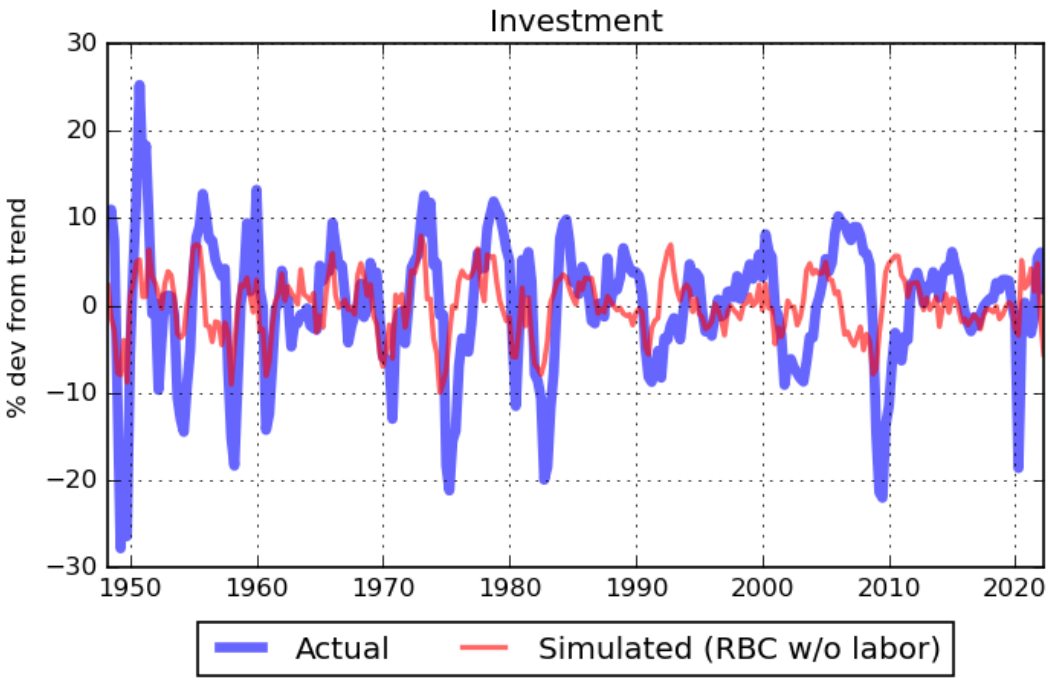

Table 3: **Standard deviations.** Actual data and simulated data. Units are percent deviations from trend. Source: FRED.

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| Consumption | 1.34 | 0.93 | 0.22 |
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Table 4: **Correlations with GDP.** Actual data and simulated data. Source: FRED.

| | Actual Data | Solow | RBC w/o Labor |
|---|---|---|---|
| Consumption | 0.81 | 1.0 | 0.61 |
| Investment | 0.84 | 1.0 | 0.99 |

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- **Next**: Add a labor-leisure tradeoff to the household's problem.

# References

**Prescott, Edward C.**, "Theory Ahead of Business Cycle Measurement," *Federal Reserve Bank of Minneapolis Quarterly Review*, Fall 1986, *10* (4), 9–22.